# REPORT OF THE PORTFOLIO COMMITTEE ON ENERGY AND POWER DEVELOPMENT

## ON THE MANUFACTURING AND SUPPLY OF TRANSFORMERS IN ZIMBABWE

## FIFTH SESSION-NINTH PARLIAMENT

***Presented to Parliament 2022***

**(S.C. 44, 2022)**

**ORDER OF APPOINTMENT**

**STANDING ORDER NO. 18**

- At the commencement of every Session, there must be as many Committees to be designated according to government portfolios as the Committee on Standing Rules and Orders may deem fit.
- Each Select Committee must be known by the portfolio determined for it by the Committee on Standing Rules and Orders.

**TERMS OF REFERENCE**

**STANDING ORDER NO. 21**

**Subject to these Standing Orders, a Portfolio Committees must-**

- examine expenditure administration and policy of government departments and other matters falling under their jurisdictions as Parliament may, by resolution determine;
- consider and deal with all Bills other than the Constitutional Bill and Statutory Instruments or other matters which are referred to them by or under a resolution of the House or by the Speaker;
- consider or deal with an Appropriation of Money Bill or any aspect of an Appropriation or Money Bill referred to them by these Standing Orders or by resolution of the House;
- monitor, investigate, enquire into and make recommendations relating to any aspect of the legislative programme, budget, policy or any other matter it may consider relevant to the government department falling within the category of affairs assigned to them, and may for that purpose consult and liaise with such a department; and
- consider or deal with all international treaties, conventions and agreements relevant to them, which are from time to time negotiated, entered into or agreed upon.

## 1.0 INTRODUCTION

1.1 The Ministry of Energy and Power Development through its various parastatals has a mandate to provide adequate and sustainable energy in the country. The National Development Strategy 1 also puts emphasis on improving energy access in Zimbabwe. However, some parts of the country continue to face electricity connectivity challenges due to the shortages of transformers, even though, there are several reputable and established transformer manufacturers in Zimbabwe. Accordingly, the Portfolio Committee on Energy and Power Development was prompted to conduct an inquiry into the manufacturing and supply of transformers in Zimbabwe. This was triggered by the reported shortages of transformers across the country, with Zimbabwe Electricity Transmission and Distribution Company submitting a total of 1900 faulty transformers.

## 2.0 OBJECTIVES

a) To assess whether local transformer manufacturers have the capacity to meet the ever-increasing demand for transformers in the country;

b) To appreciate challenges faced by local transformer manufacturers; and

c) To come up with recommendations for improved manufacturing and supply of transformers in the country.

## 3.0 METHODOLOGY

3.1 This report is informed by written submissions from Zimbabwe Electricity Supply Authority (ZESA). It is also informed by oral submissions from officials from Zesa Enterprises (ZENT), Hawker Siddeley Engineering (South Wales), Nical Transformers and SE Electrical Engineers. The oral evidence meetings were held on 24 March, 2022 and 05 May, 2022.

3.2 In order to get official and technical insight into the capacity of local manufactures to manufacture and supply transformers and challenges being faced in the industry, the Committee conducted verification visits to selected transformer manufacturing plants from 18 May to 19 May, 2022. The Committee visited the following plants;

| **District** | **Plant** |
|---|---|
| Harare | Zesa Enterprises (ZENT) |
| Harare | Hawker Siddeley Engineering (South Wales) |
| Harare | Nical Transformers |

| | |
|---|---|
| Harare | SE Electrical Engineers |

## 4.0 COMMITTEE FINDINGS

### 4.1 Historical background

4.1.1 Prior to unbundling of Zimbabwe Electricity Supply Authority (ZESA), transformer repairs were done under the Production and Services Business Unit which had a capacity to repair only 40 transformers per month and there was no capacity to manufacture. After unbundling of ZESA in 2004, a new mandate was given to the successor business unit, Zesa Enterprises (ZENT) to sustainably support the electricity industry with electrical goods and services such as transformers. In order to fulfil this new mandate, a decision was taken to capacitate the business in design, technical skills, machinery and organizational capacity through a technology transfer partnership with an Indian Company called PME. The technology transfer agreement ran from 2005 to 2017 and enabled ZENT to acquire machinery and equipment with a manufacturing capacity of 3000 transformers per annum through credit. However, the plant has aged and become unreliable as it was overtaken by new modern technologies.

4.1.2 The local private manufacturers such as South Wales explained that prior to the unbundling of ZESA, the transformer manufacturing industry was self-regulated. This promoted competitive pricing and strengthened the manufacturing capacity of these companies.

### 4.2 Growth potential of the transformer manufacturing market

4.2.1 According to submissions from ZESA, access to electricity in the country was estimated at 43% of the population. Therefore, the national objective was to increase the access to electricity to 80% by 2030, in line with National Vision 2030, which aims to have an upper middle-income society by 2030. ZESA as the principal energy supplier also has a vision of total electrification by 2030. The accelerated electrification requires massive expansion of the backbone infrastructure for transmission and distribution. This expansion calls for an increased output of transformers beyond the existing capacity of 3000 per annum.

4.2.2 At the time of the enquiry, a total of 305 000 households required electrical power connections. The demand for power connectivity and transformers was expected to increase due to 250 000 houses expected to be constructed under the government's accelerated housing delivery program.

**4.3 State of affairs in the supply of transformers in Zimbabwe**

4.3.1 Zimbabwe has reputable and established transformer manufactures such as ZENT, Nical Transformers and South Wales, among others. The Committee learnt that local manufacturers had a capacity to design, manufacture, test and supply an average of 100-250 transformers per month subject to availability of foreign currency and competitively priced key raw materials.

4.3.2 Some of the local manufacturers such as ZENT also have capacity to manufacture other electricity infrastructure such as concrete poles, substation, line accessories, LV distribution boards, metering cubicles, protection, control panels and gang nails, which are vital for electricity connection in the country.

4.3.3 During the tour of ZENT premises, it was stated that Rural Electrification Fund (REF), Zimbabwe Electricity Transmission and Distribution Centre (ZETDC) and Electricidade de Moçambique (EDM) contracted it to manufacture a total of 6100 transformers. However, ZENT only managed to deliver 3716 transformers due to foreign currency challenges. The utility had set a target to manufacture 1982 transformers in 2022 against the order balance of 2384 transformers. At the time of the visit by the Committee in May 2022, the utility had only manufactured 161 transformers since January 2022 due to shortages in foreign currency and delays in deliveries of materials caused by COVID-19 pandemic measures.

4.3.4 During the oral evidence meeting with private local manufacturers, the Committee learnt that local manufacturers such as Hawker Siddeley Engineering (South Wales), Nical Transformers and SE Electricals Engineers could not export their locally manufactured transformers due to high cost of production which makes their exports uncompetitive. Moreover, the cost and duty charged on imported materials made locally manufactured transformers more expensive when compared to transformers from other countries particularly China and India. As a result, locally manufactured transformers could not be competitive in the global industry.

4.3.5 The local manufacturers also indicated that their transformers were also more costly in the local industry compared to imported transformers. As such, an imported 25KvA aluminium transformer was priced around US$1000-US$1800 whilst locally produced were around US$3300. Consequently, local clients preferred to purchase imported transformers even though they could not last longer compared to the locally produced transformers.

4.3.6 During the tour of private local transformer manufacturing plants, the Committee learnt that most local transformer manufacturers were more occupied with repairing

transformers than manufacturing as majority of transformers in the country were imported and often broke down. Even though all imported or locally produced transformers undergo tests at ZENT facilities, the local manufacturers explained that the test could not determine the life span of a transformer which made it possible for any transformer to pass the test whilst it is not durable.

4.3.7 The local manufacturers highlighted that the influx of cheap imported transformers of poor quality, particularly from India and China had crippled the local manufacturing industry. The local manufacturers feared that continually importing transformers would eventually lead to the closure of the local industry. This would enable Indian and Chinese transformer manufacturing companies to take over the industry and charge high prices as they do in other countries. Whilst churning out substandard transformers which frequently needs replacement.

4.3.8 Some of the manufacturers such as SE Electricals Engineers also indicated the need for Government to consider allowing waivers for local manufacturers to be able to participate in the auction of foreign currency using loans. This would allow them to improve their manufacturing capacity and produce more transformers to support the energy sector.

### 4.4 Distribution of transformers in Zimbabwe

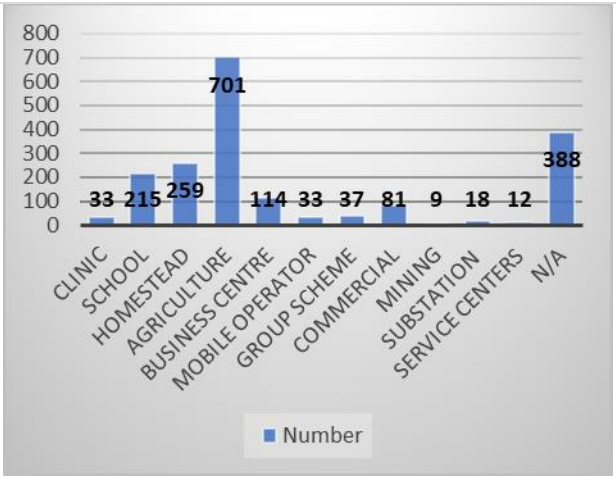
4.4.1 According to the submissions from ZESA, ZETDC was negatively affected by shortages of transformers and other materials. The procurement of transformers had been in progress since 2020. However, very few transformers were delivered by suppliers due to shortages of foreign currency and Covid 19 restrictions. As a result, the backlog of transformers remains huge and this continues to incapacitate the utility to distribute transformers to areas in need. Therefore, the utility was prioritizing essential service providers such as health institutions, agriculture, security sector, water works, industry and commerce, instead of the first come first serve basis.

### 4.5 Replacement of faulty transformers in the country

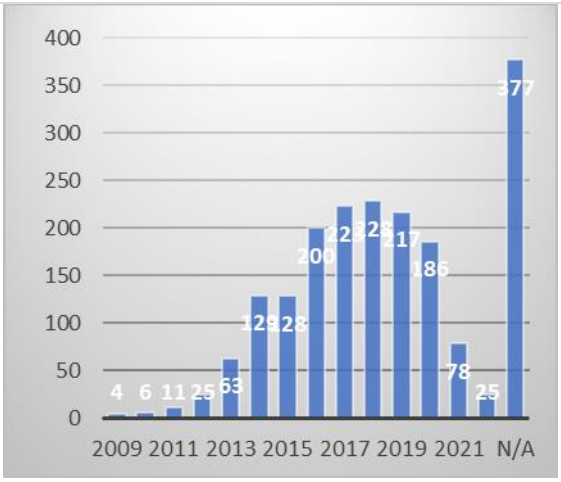
4.5.1 The submissions from ZESA highlighted that around 1900 transformers broke down since 2009 to 2022. 946 of them were from the Northern region, 128 from the Southern region, 124 from the Western region and 701 from the Eastern region.

4.5.2 The below graphs depict the summary of the information shared by ZESA on sectors that had transformers that broke down and the numbers of transformers that were not working. The graphs further show the years within which transformers broke down as well as the number of transformers that broke down each year. Conclusively, the graphs show the causes of the breakdown of transformers in the various sectors.

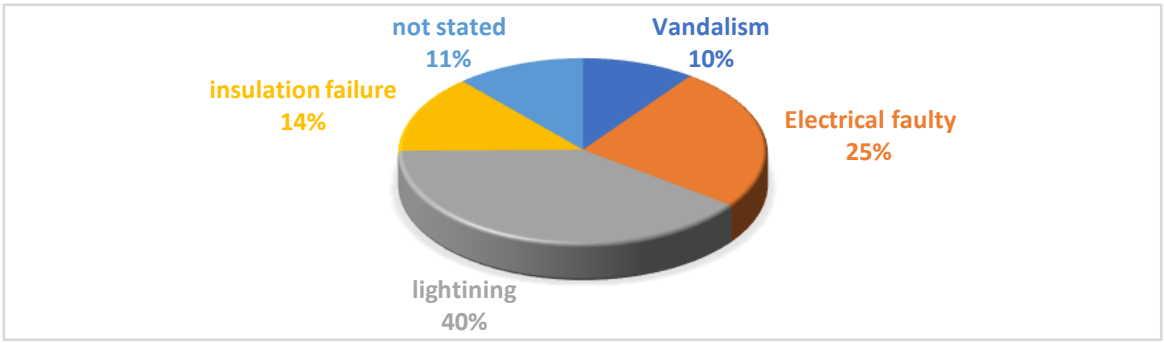
**Figure 1: Analysis of faulty transformers in Zimbabwe**




**Distribution of faulty transformers by sector**




**Year of failure of transformers**




**Causes of failure**

4.5.3 From the above graphs, agriculture was seen to be the sector with most faulty transformers accounting to 36.9%, followed by homesteads (13.6%) and schools (11.2%). The Committee learnt that lightning contributed to 40% of transformer failure within the four regions of Zimbabwe. Electrical fault which might be due to ageing infrastructure contributed to 25% of total transformer failure in the country. Insulation failure, which might be subject to human error during winding or machine error especially on imported transformers contributed to 14% of the transformer failure in Zimbabwe. Whereas, vandalism (10%) was the least cause of the breakdown of transformers. The Committee also learnt that there was a sudden increase in the rate of failure of transformers from 2012 compared to 2011 and beyond.

## 4.6 Investment opportunities in the industry

4.6.1 The Committee learnt that the transformer manufacturing sector was heavily reliant on imported materials. As such, there was an opportunity for investors to set up plants to manufacture raw materials such as porcelain and bushings instead of competing to supply

transformers to the country. Such investments would also reduce the country's importation bill and contribute to employment creation.

4.6.2 It was also highlighted that investing in aluminum technology could be a cheaper alternative to curb the rampant vandalism of transformers but are not durable compared to copper.

## 5. CHALLENGES AFFECTING TRANSFORMER DELIVERY

### 5.1 *Shortages of foreign currency*

It was indicated that around 80% of transformer raw materials were imported and as such foreign currency was required. However, the industry was affected by shortages in foreign currency which delayed procurement of materials. This was further worsened by failure to secure lines of credit.

### 5.2 *Increment in the prices of raw materials*

It was highlighted that the sector was affected by the continuous increase in prices of raw materials, as dictated by world market prices due to Covid 19 pandemic. Recent events in Eastern Europe also influenced a new wave of price upsurges and destabilization of supply chains. As a result, suppliers were hesitant to do business in a volatile environment which further delayed the importation of raw materials in the country.

### 5.3 *Obsolete equipment*

ZENT expressed concern that it was operating with aged and obsolete equipment that timeously required spares that were unavailable and this reduced its capacity to manufacture transformers as well as its competitiveness in the industry.

### 5.4 *High Import Duty*

Local manufacturers bemoaned that imported raw materials were charged duty thereby increasing the local manufacturing costs and reduced the competitiveness of the industry in the global market. It was indicated that importation of cheap and poor-quality transformers from China and India was crippling the local manufacturing industry. In addition, they expressed concern on the absence of support for local manufacturers which had been demonstrated by liberalizing the industry to allow locals to compete with foreign suppliers with more efficient technologies.

### 5.5 *Monopoly of ZENT*

The local manufacturers expressed concern with the role played by ZENT as both a competitor and regulator who certified all transformers at high costs. The local manufacturers also highlighted that they have capacity to test transformers at their plants and yet they were still required to go for a second testing at ZENT. ZENT was currently said to be charging high prices as much as US$35 for a 10kVA and US$780 for a 2000kVA and daily storage rates of US$20. Such costs were incorporated in the final product cost which ends up increasing prices of transformers. Private players were worried that ZENT may not charge itself thus it does not affect its pricing of transformers. They stated that centralisation of testing in Harare burdens other players who are outside Harare as they have to carry their transformers to ZENT and incur further production costs. Hence, they suggested that there was need for an independent body that would regulate the industry. They also recommended for decentralization of ZENT's testing and certification facilities to cater for areas outside Harare.

### 5.6 *Awarding of tenders to briefcase companies*

The Committee was informed that the local manufacturers were saddened by the awarding of tenders to companies without the capacity to design, manufacture, test and supply. These companies would end up subcontracting other capacitated local manufacturers to deliver the services on their behalf.

### 5.7 *Copper recovery programme*

The local manufacturers also expressed their dismay that they were not being given an opportunity to participate in the copper recovery programme. The Committee was informed that CAFCA, a cable manufacturing company, was the only local company that had been granted the permission to recover copper components from faulty transformers and yet the copper was also important in the transformer manufacturing industry.

## 6. COMMITTEE OBSERVATIONS

The Committee made the following observations;

a) ZENT acts as a regulator and a competitor in the transformer manufacturing sector thereby enjoying unfair advantage over other local manufacturers.

b) ZENT was inadequately capacitated to locally manufacture transformers and other electricity equipment.

c) There is no due diligence being done to ensure that all imported transformers come from

reputable and certified companies with warrant.

d) The technology at ZENT is outdated which is affecting its efficiency and reliability in manufacturing and delivering transformers.

e) There is no policy framework from Government to promote the buying of locally produced transformers in place of importing substandard and outmoded transformers.

f) ZENT's testing and certification facilities are not decentralized which makes it costly and difficult for local transformer manufactures who operate outside of Harare.

g) The recovery of copper components from faulty transformers is only done by CAFCA and excludes other local transformer manufacturers who can make use of the components.

## 7. COMMMITEEE RECOMMENDATIONS

In light of the above observations, the Committee recommended the following;

a) The Ministry of Finance and Economic Development should introduce waivers for local transformer manufacturers to be able to participate in the auction of foreign currency using loans, as it will enable them to improve their capacity as manufacturers by March 2023.

b) The Ministry of Energy and Power Development should ensure that ZENT is adequately capacitated to locally manufacture transformers and other electricity equipment by January 2023.

c) Government should ensure that imported transformers should come from reputable and certified companies with warrant, with immediate effect.

d) Government should restrict importation of transformers once the country manages to clear the outstanding balance of transformers and invest in capacitating other private local manufacturers to meet ever increasing demand of transformers and other electricity equipment.

e) Zesa Enterprises should engage another technology transfer partner from a developed country that can assist with modern technology to improve its efficiency and reliability by April 2023.

f) Government should introduce measures that promote the buying of locally produced transformers in place of importing substandard and outmoded transformers by March 2023.

g) Zesa Enterprises should decentralize its testing and certification facilities to bring convenience to local transformer manufactures who operate outside of Harare by March 2023.

h) The Ministry of Energy and Power Development should ensure that other interested local transformer manufacturers are permitted to participate in the copper recovery programme by January 2023.

## 8. CONCLUSION

The energy sector is a vital part of the economy's nervous system which enables progress in other sectors as well as the country to execute other Sustainable Development Goals. However, the current environment is an impediment to the vision of the country to become a self-reliant and a net exporter of electricity as Zimbabwe moves toward a middle-income economy by 2030.In that regard, the Committee is expectant that the relevant authorities will consider the above recommendations proffered for the purposes of establishing a vibrant energy sector that responds to the growing economy and targeted vision.